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Performance Analysis of L-Branch Equal Gain Combiners in Equally Correlated Rayleigh Fading Channels

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***Abstract*—Theoretical performance results for L-branch ($L \geq 3$) coherent equal-gain combining (EGC) in correlated fading channels are not known. This letter develops a novel approach for performance analysis of L-branch EGC in equally correlated Rayleigh fading channels. Such channel gains can be transformed into a set of conditionally independent channel gains. The cumulative distribution function (cdf) of the EGC output signal-to-noise ratio (SNR) is, therefore, derived. The symbol error rate (SER) of different modulation schemes with EGC in equally correlated Rayleigh fading channels is evaluated. Numerical results that illustrate the effects of equally correlated fading on the SER performance of EGC are also provided.**

***Index Terms*—Cumulative distribution function, diversity, equal-gain combining (EGC), equally correlated Rayleigh fading channels, symbol error rate (SER).**

I. Introduction

EQUAL-GAIN combining (EGC)is of practical interest because it offers performance close to the optimal maximal-ratio combining (MRC) technique but with greater simplicity (see [1] for an extended discussion on EGC and MRC performance difference). The error rates and outage performance of L-branch EGC in *independent* fading channels are well known [2][3][4][5][6]. However, to the best of the authors' knowledge, theoretical performance results for L-branch EGC in *correlated* fading channels do not exist for $L > 2$. This problem is important from both a theoretical and practical standpoint, as residual correlations among diversity branches exist due to several reasons (e.g., insufficient antenna spacing). Unfortunately, the existing results for EGC performance in correlated fading hold for the dual-branch case only. Mallik *et al.* [7] derive the bit error rate (BER) for coherent detected binary signals with dual-branch EGC in correlated Rayleigh fading channels. Tellambura and Annamalai [8] derive the characteristic function (chf) of the EGC output and utilize an infinite series representation for the complementary error function to obtain the BER. In [9], an integral representation for the Gaussian probability integral is used to derive the BER.

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In this letter, we develop a novel approach to analyze the performance of L-branch EGC in equally correlated Rayleigh fading channels. The novel insight of this letter is that a set of independent Gaussian random variables (RVs) can be linearly combined to form a set of equally correlated complex Gaussian RVs. Using this insight, we translate the problem of EGC performance in equally correlated Rayleigh fading to the problem of EGC performance in a conditionally independent Rician fading environment. This reformulation allows us to extend the known results for independent Rician fading to analyze the L-branch EGC performance in equally correlated Rayleigh fading.

This letter is organized as follows. Section II develops a new representation for the equally correlated Rayleigh channel gains. Section III derives the cdf of the EGC output in such channels. Section IV evaluates the SER of various digital modulations with EGC. Numerical results in Section V illustrate the impact of fading correlation on EGC performance.

II. Representation of Channel Gains

We represent the Rayleigh envelopes using a set of zero-mean complex Gaussian RVs given by

$$G_k = \alpha_k \left[(\sqrt{1-\rho} X_k + \sqrt{\rho} X_0) + i(\sqrt{1-\rho} Y_k + \sqrt{\rho} Y_0) \right] \tag{1}$$

for $k = 1, \ldots, L$, where $i = \sqrt{-1}$, $0 \leq \rho \leq 1$, α_k is a set of constants and X_k and Y_k, $(k = 0, 1, \cdots, L)$, are two sets of independent zero-mean and unit-variance Gaussian RVs. That is, for any $j, k \in \{0, \ldots, L\}$, $E(X_k Y_j) = 0$ and

$$E(X_k X_j) = E(Y_k Y_j) = \begin{cases} 1, & k = j \\ 0, & k \neq j \end{cases} \tag{2}$$

where $E(x)$ denotes the average of x.

We may write the G_k in terms of polar coordinates as $G_k = R_k \exp(i\theta_k)$, $(k = 1, 2, \cdots, L)$, where $R_k = |G_k|$ is the amplitude of the G_k. Since G_k is a set of complex Gaussian RVs with zero means, R_k is a set of Rayleigh envelopes with mean square $E(R_k^2) = 2\alpha_k^2$.

Equation (1) allows two correlation models. The correlation (covariance) between diversity branches k and j $(k \neq j)$ can be

$$E\left(G_k G_j^*\right) = \begin{cases} \alpha_k \alpha_j \rho, & \text{Model 1} \\ \alpha^2 \rho, & \text{Model 2} \end{cases} \tag{3}$$

where $\alpha_1 = \alpha_2 = \cdots = \alpha_L = \alpha$ for model 2. Model 1 allows us to study the impact of branch gain imbalance and correlation. In model 2, the amount of correlation between any two branches is the same, whereas in model 1, the amount of correlation is proportional to the product $\alpha_k \alpha_j$. Nevertheless, for

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both models, the cross-correlation coefficient between any G_k and G_j $(k \neq j)$ is constant as

$$\frac{E\left(G_k G_j^*\right)}{\sqrt{E\left(|G_k|^2\right) E\left(|G_j|^2\right)}} = \rho. \tag{4}$$

For this reason, we refer to both the models as being equally correlated. The relationship between the power correlation (i.e. the correlation between G_k and G_j) and the envelope correlation (i.e. the correlation between the R_k and R_j) is given by an elliptic integral [10, eq. (1.5–26)]. Thus, we can readily represent a set of equally correlated Rayleigh envelopes with a specified value of envelope correlation.

III. Derivation of EGC Output CDF

Using the new representation for the channel gains (1), we derive the cdf of the L-branch EGC output SNR. We assume the receive signals at different branches to be equally correlated with each other. The noise components at different branches are assumed to be identical and independent of the signal components and uncorrelated with each other.

In EGC, the received signals are first co-phased and then summed to form the resultant output. The instantaneous SNR at the output can be written as

$$\gamma_{egc} = \frac{(R_1 + R_2 + \cdots + R_L)^2 E_s}{L N_0} \tag{5}$$

where L denotes the diversity order, N_0 is the noise power spectral density per branch and E_s is the energy of transmitted signal. We define the branch imbalance coefficient as

$$\frac{E\left(R_k^2\right)}{E\left(R_1^2\right)} = \frac{\alpha_k^2}{\alpha_1^2} = \lambda_k, \quad k = 1, 2, \ldots, L. \tag{6}$$

We consider X_0 and Y_0 to be fixed $(X_0^2 + Y_0^2 = U)$ and let $V_k = R_k/\alpha_1$. Then $V = V_1 + V_2 + \cdots + V_L$ is a sum of L independent Rician RVs with the Rice factor and the average power given by

$$K_k = \frac{\rho u}{2(1-\rho)}, \tag{7a}$$

$$\Omega_k = \left[2(1-\rho) + \rho u\right] \lambda_k, \quad k = 1, \cdots, L. \tag{7b}$$

The SNR of the kth branch is

$$\overline{\gamma}_k = \frac{E_s}{N_0} E\left(R_k^2\right) = \frac{2\alpha_k^2 E_s}{N_0}. \tag{8}$$

Evaluating the cdf of the EGC output for a fixed $x_0^2 + y_0^2 = u$, we obtain the following conditional output cdf:

$$\begin{aligned} F_{\gamma_{egc}}\left(y|_{x_0^2+y_0^2=u}\right) &= \Pr\left(\gamma_{egc} \leq y|_{x_0^2+y_0^2=u}\right) \\ &= F_V\left(\sqrt{\frac{2Ly}{\overline{\gamma_1}}}\bigg|_{x_0^2+y_0^2=u}\right) \end{aligned} \tag{9}$$

where $F_V(x|_{x_0^2+y_0^2=u})$ is the cdf of V given $x_0^2 + y_0^2 = u$, which can be computed using a convergent infinite series [2], [4]

$$F_V\left(x|_{x_0^2+y_0^2=u}\right) \cong \frac{1}{2} - \frac{2}{\pi} \sum_{n=1,odd}^{\infty} \frac{A_n \sin(\theta_n)}{n} \tag{10}$$

where A_n and θ_n are conditioned on $x_0^2 + y_0^2 = u$ [4]

$$A_n = \prod_{k=1}^{L} \sqrt{E^2\left[\cos\left(\frac{2\pi n V_k}{T}\right)\right] + E^2\left[\sin\left(\frac{2\pi n V_k}{T}\right)\right]} \tag{11a}$$

$$\theta_n = \prod_{k=1}^{L} \tan^{-1}\left(\frac{E^2\left[\sin\left(\frac{2\pi n}{T}\left(V_k - \frac{x}{L}\right)\right)\right]}{E^2\left[\cos\left(\frac{2\pi n}{T}\left(V_k - \frac{x}{L}\right)\right)\right]}\right) \tag{11b}$$

and T is the period of squarewave defined in [2]. Given $x_0^2 + y_0^2 = u$, the pdf of V_k can be obtained as [11]

$$\begin{aligned} p_{V_k}\left(x|_{x_0^2+y_0^2=u}\right) = &\frac{x}{\lambda_k(1-\rho)} \exp\left(-\frac{\rho u \lambda_k + x^2}{2\lambda_k(1-\rho)}\right) \\ &\times I_0\left(\frac{x}{1-\rho}\sqrt{\frac{\rho u}{\lambda_k}}\right), \quad x \geq 0. \end{aligned} \tag{12}$$

Hence, we may compute $E[\cos(2\pi n V_k/T)]$ and $E[\sin(2\pi n V_k/T)]$ given $x_0^2 + y_0^2 = u$ using [4]

$$\begin{aligned} &E\left[\cos\left(2\pi n \frac{V_k}{T}\right)\right] \\ &= \exp\left(-\frac{\rho u}{2(1-\rho)}\right) \sum_{j=0}^{\infty} \left(\frac{\rho u}{2(1-\rho)}\right)^j \frac{1}{j!} \\ &\quad \times {}_1F_1\left(j+1, \frac{1}{2}, \frac{2\pi^2 n^2 (1-\rho)\lambda_k}{T^2}\right) \end{aligned} \tag{13a}$$

$$\begin{aligned} &E\left[\sin\left(2\pi n \frac{V_k}{T}\right)\right] \\ &= \frac{2\pi n \sqrt{2(1-\rho)\lambda_k}}{T} e^{-\frac{\rho u}{2(1-\rho)}} \sum_{j=0}^{\infty} \frac{\Gamma\left(j+\frac{3}{2}\right)}{(j!)^2} \left(\frac{\rho u}{2(1-\rho)}\right)^j \\ &\quad \times {}_1F_1\left(j+\frac{3}{2}, \frac{3}{2}, -\frac{2\pi^2 n^2 (1-\rho)\lambda_k}{T^2}\right) \end{aligned} \tag{13b}$$

where ${}_1F_1(a, b, x)$ is the confluent hypergeometric function defined as [12, eq. (13.1.2)].

Notice that $X_0^2 + Y_0^2$ is chi-square distributed with two degrees of freedom and its pdf can be found [11]

$$p(u) = \frac{1}{2} e^{-\frac{u}{2}}, \qquad u \geq 0. \tag{14}$$

Averaging the conditional cdf (9) over the distribution of U (14), we finally obtain the cdf for L-branch EGC in equally correlated Rayleigh fading channel

$$F_{\gamma_{egc}}(y) = \frac{1}{2} \int_0^{\infty} F_V\left(\sqrt{\frac{2Ly}{\overline{\gamma_1}}}\bigg|_{x_0^2+y_0^2=u}\right) e^{-\frac{u}{2}} du \tag{15}$$

where $F_V(x)$ can be computed using (13) and (11) with (10). This is a novel result which enables the evaluation of the outage of L-branch EGC systems in equally correlated Rayleigh fading channels. For the independent case, our new result (15) reduces to the well-known result [2, eq. (5)]. Equation (15) can be readily evaluated via numerical quadrature techniques with high accuracy (details omitted for brevity).

IV. Performance Analysis

Following the same approach, we may evaluate the average SER of various digital modulations with EGC in equally correlated Rayleigh fading channels. We assume that channel fading is nonselective and changing slowly enough so that the channel

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parameters remain constant for the duration of the signaling interval.

Recall that when X_0 and Y_0 are fixed ($X_0^2 + Y_0^2 = U$), R_k is a set of independent Rician RVs with the Rician factor $K_k = \rho u/2/(1-\rho)$ and the average power $\Omega_k = [2(1-\rho) + \rho u]\alpha_k$, $(k = 1, \cdots, L)$. Due to Annamalai *et al.* [13], we can readily obtain the conditional SER for a fixed $x_0^2 + y_0^2 = u$

$$P_s(u) = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \frac{\Psi\left(\tan\zeta|_{x_0^2+y_0^2=u}\right)}{\sin(2\zeta)} d\zeta \tag{16}$$

where

$$\Psi\left(\omega|_{x_0^2+y_0^2=u}\right) = \mathrm{Real}\left[\omega G(\omega)\phi_r^*\left(\omega|_{x_0^2+y_0^2=u}\right)\right] \tag{17}$$

where $\mathrm{Real}(x)$ denotes the real part of x and $*$ denotes the conjugate operation. $G(\omega)$ is the Fourier transform (FT) of generic conditional error probability which can be found in [13] and $\phi_r(\omega|_{x_0^2+y_0^2=u})$ is the chf of the square root of output SNR $(\sqrt{\gamma_{egc}})$ when $x_0^2 + y_0^2 = u$ is fixed, which is given by [13]

$$\phi_r\left(\omega|_{x_0^2+y_0^2=u}\right) = \prod_{k=1}^{L}\left[\frac{e^{-\frac{\rho u}{2(1-\rho)}}}{\pi}\int_0^{\pi} e^{\left(i\frac{\omega\xi_k}{2}+\sqrt{\frac{\rho u}{1-\rho}}\frac{\cos\theta}{2}\right)^2} \times D_{-2}\left(-i\omega\xi_k - \sqrt{\frac{\rho u}{1-\rho}}\cos\theta\right)d\theta\right] \tag{18}$$

where $\xi_k = \sqrt{(1-\rho)\overline{\gamma_k}/2/L}$ and $D_{-2}(x)$ is the parabolic cylinder function defined as [14, eq. (9.240)].

Averaging the conditional SER (16) over the distribution of U (14), we obtain the average SER as

$$\overline{P_s} = \frac{1}{\pi}\int_0^{\infty}\int_0^{\frac{\pi}{2}} \frac{\Psi\left(\tan\zeta|_{x_0^2+y_0^2=u}\right)}{\sin(2\zeta)} d\zeta e^{\frac{-u}{2}} du. \tag{19}$$

Therefore, we may evaluate the average SER of various digital modulations with EGC using [13, eq. (11, 13, 16, 17, 20)] and (18) with (19).

V. Numerical Results and Conclusion

Fig. 1 shows the impact of the power correlation coefficient ρ on the performance of binary differential phase-shift keying (DPSK) with balanced L-branch EGC. The case of $L = 1$ represents a situation with no diversity. As expected, the diversity gain can still be achieved even in highly correlated fading channels. The maximum additional diversity gain is achieved with dual-branch diversity. With increasing L, the additional diversity gain diminishes. This is more pronounced at higher SNRs. We also observe that, as ρ increases, the performance of DPSK degrades more rapidly with larger L.

In conclusion, we have developed a new representation for equally correlated Rayleigh channel gains and this enables the performance analysis of EGC in correlated fading. Numerical results show that diversity gains can still be achieved in correlated Rayleigh fading channels. Our representation can also be used to analyze the performance of L-branch selection combining (SC) and generalized SC (GSC) schemes in equally correlated Rayleigh fading channels. These results will be reported in a forthcoming paper. It should be emphasized that our approach holds only for equally correlated fading and whether our approach can be generalized to handle other correlation models is an on-going research problem.

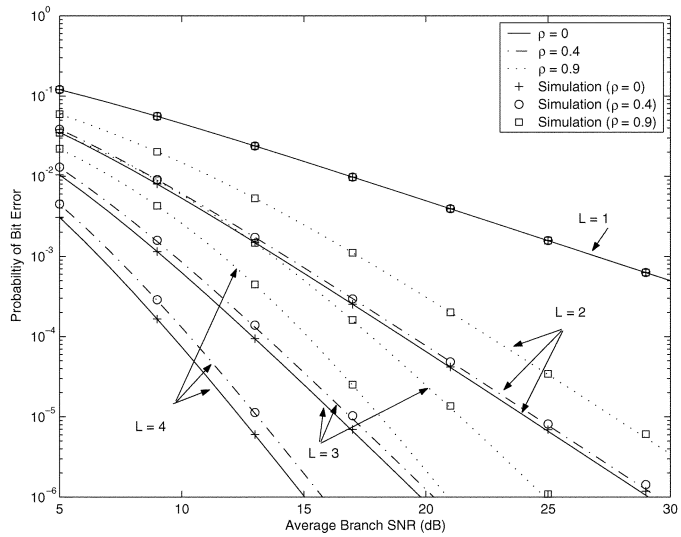

Fig. 1. Average BER of DPSK with EGC in equally correlated Rayleigh fading channels; $\rho = 0$, 0.4, 0.9.

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